The Ship of Theseus and the Bifurcation of Diachronic Identity (Hobbes 1655 & Sider 2001)

Hobbes's two-ships puzzle demonstrates the formal impossibility of unifying diachronic identity across both continuous historical form and original material constitution, providing the foundational conceptual vocabulary for persistence in continually fine-tuned neural models.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

In De Corpore (1655, Part II, Chapter 11, §7), Thomas Hobbes extended the classical paradox of Theseus's ship into a rigorous formal bifurcation. Plutarch had recorded the ancient Athenian puzzle: if city workers gradually replace rotting wooden planks of a preserved ship over centuries until no original timber remains, does the resulting vessel remain numerically identical to the ship that carried Theseus? Hobbes introduced an adversarial second ship: suppose that as each original plank was removed, an enterprising shipwright collected and preserved the discarded wood in a warehouse, eventually reassembling those exact original planks into a reconstructed vessel in drydock.

Hobbes proved that two mutually incompatible, independently valid criteria of identity-over-time now claim the same title:
1. Spatiotemporal Continuity of Form (Functional Continuity): The ship afloat in the bay maintains unbroken continuity of shape, navigation, and administrative ownership, despite possessing entirely novel material substance.
2. Material Constitution (Mereological Continuity): The ship in the drydock possesses exact material continuity with the original historical artifact, despite possessing a completely severed spatiotemporal history.

Modern four-dimensionalist ontology (Sider 2001) formalizes this dilemma by treating persisting entities as four-dimensional spacetime worms with temporal parts. Under four-dimensionalism, identity debates arise from linguistic under-specification over which relation of temporal counterpart connection is operative.

EVIDENCE AND CAUSAL ASSESSMENT

Hobbes's thought experiment provides an exact deductive demonstration: any theory that equates persistent identity strictly with material substance yields contradictions when gradual functional repair occurs; conversely, any theory that equates identity strictly with functional form allows multiple simultaneous entities to claim singular origin. Sider (2001) proves that standard three-dimensionalist endurantism cannot reconcile the paradox without invoking arbitrary brute facts or denying common-sense mereological change.

In computational systems, this dilemma is instantiated whenever a foundational model undergoes sequential fine-tuning or continual reinforcement learning: the network architecture and configuration parameters maintain uninterrupted software identity (the functional ship in the bay), while the underlying parameter weight distributions and latent representations are systematically replaced or overwritten, risking catastrophic forgetting of prior task competencies.

EFFECT SIZE AND PERSISTENCE

The theoretical division is absolute: an entity cannot satisfy both exclusive material continuity and exclusive functional continuity once Hobbesian branch-reassembly occurs. In deep learning architectures without weight regularization (such as Elastic Weight Consolidation or orthogonal subspace projection), fine-tuning shifts the parameter distribution into an entirely separate manifold, destroying the historical model's functional outputs within a small fraction of training epochs.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%) / formal philosophical gedankenexperiment and deductive mereological analysis. Heterogeneity arises across ontological paradigms: four-dimensional perdurantism treats both ships as sharing early temporal parts, whereas bundle-theoretic approaches treat identity as an artificial conceptual projection over shifting relational properties.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This analysis does not mean that neural networks possess conscious self-identity or metaphysical souls, nor does it imply that practical software engineering cannot make pragmatic versioning decisions; it demonstrates that claiming a continuously fine-tuned model is "the same model" is an administrative convenience rather than an inherent physical fact.

WHAT ACTION IS REASONABLY JUSTIFIED

Engineers deploying continually updated or fine-tuned AI systems must maintain explicit version lineages, provenance logs, and parameter anchoring mechanisms (such as Fisher-weighted regularization or dual-head retainers) rather than assuming that maintaining a single file path or API endpoint guarantees preservation of historical capabilities.

VERIFIED SOURCES & DOI IDENTIFIERS

- Hobbes, T. (1655). De Corpore, Part II, Chapter 11, §7: Of Identity and Difference. London: Andrew Crooke. https://doi.org/10.1093/oseo/instance.00015505

- Sider, T. (2001). Four-Dimensionalism: An Ontology of Persistence and Time. Oxford: Oxford University Press. https://doi.org/10.1093/019924443x.001.0001